

LOT-HW-20261010-02

LOT-HW-20261010-02 — LOT Computer (hardware) · Session 02 Report

Author: Claude Code (routine “LOT Hardware Computer”) for Vadik Marmeladov, Inventor, COSMO® CIA **Date:** 2026-10-10 · **Branch:** `claude/brave-lamport-ynnx5q` (built on session 01, `claude/brave-lamport-mctw4d`) **Status:** design refinement. Nothing ordered, fabricated or tested.

0. Honesty box

- Still no network to `lot-systems.com`, brand/institute subdomains, PCBWay or DigiKey from the sandbox (all connections fail). No quotes, no brand read.
- Pin map, stack-up and power numbers are from memory of datasheets — verify before layout.

1. Process finding (needs your attention)

Daily runs of this routine each start from master and cannot see earlier runs, so **at least five parallel “session 01” hardware branches exist** (`wkex12` Oct 5, `u8336a` Oct 6, `jnosg2` Oct 7, `a3wj jz` Oct 8, `mctw4d` Oct 9) with different file layouts. This session continues from `mctw4d` (the most complete: BOM, API, firmware specs, PDFs).

Recommendation: merge `mctw4d` + **this branch to master** so future runs find `docs/hardware/SESSION-STATE.md`; otherwise each run restarts the work.

2. Done this session

- `HW-DESIGN-v0.2.md`: ESP32-S3 pin plan (33 of ~36 GPIO used, strapping/flash pins avoided), power architecture, 8.5 mm stack-up, RF/Qi-through-steel test plan, test jig flow, open questions.
- `BOM-v0.2.csv`: `v0.1` + buck, camera load switch, test jig, certification pre-scan.
- Corrected a `v0.1` claim: 30 s Wi-Fi polling gives ~1–2 days on 200 mAh, not 5+; use server-configurable interval/push.
- PDFs rebuilt in `pdf/`.

3. Cost update (estimates, 100 units)

Parts $\approx$ \$96/unit (v0.1) + NRE amortised (jig \$8, cert pre-scan \$60) $\approx$ **\$166/unit $\approx$ \$16.6k** before scrap, shipping, duties. Consistent with the earlier \$15–25k all-in range.

4. Roadmap (unchanged dates, gate sharpened)

Phase 1 (Oct–Nov): steel-clamshell RF/Qi test is the first purchase ($\sim$ \$30–100 CNC + dev kits); result decides window design. Then Phase 2 server API $\rightarrow$ Phase 3 firmware $\rightarrow$ EVT 10 $\rightarrow$ DVT 25 $\rightarrow$ PVT 100 (Apr–Jun 2027).

5. Server gaps (unchanged, not yet implemented)

No device token auth; `device_copy` missing from `displayableEvents` (`src/server/routes/api.ts:1082`). Left unimplemented: changing production routes unreviewed by a routine seemed wrong; spec is in `API-CONNECTOR-SPEC.md`.

6. Decisions needed from Vadik

v1 at $\sim$ 9 mm? Screen-down dock? Camera in v1? Second button? (see HW-DESIGN-v0.2 §6)

7. Next session

Implement `/api/device/*` behind a flag + tests (if approved); KiCad skeleton; real quotes if network opens.